

**Problem Bank to accompany Homework 3.** Each item is a short, exam-style question targeting one specific idea from the homework. The goal is *repetition* — five short passes on the same concept so the mechanics become reflexive by the midterm. Try each question on your own before consulting the solutions, which are collected at the end.

## Concept A: Leave-one-out regression via a unit dummy

**A.1** Consider data  $(x_i, y_i)$  for  $i = 1, 2, 3$ :  $(1, 2), (2, 5), (3, 4)$ . Let  $\mathbf{d} = (0, 0, 1)'$  be a dummy for observation 3.

- (a) Run OLS of  $\mathbf{y}$  on an intercept and  $\mathbf{x}$  using only observations 1 and 2. Report  $\hat{\alpha}_{(-3)}$  and  $\hat{\beta}_{(-3)}$ .
- (b) Run OLS of  $\mathbf{y}$  on an intercept,  $\mathbf{x}$ , and  $\mathbf{d}$  using all three observations. Compare the coefficients on the intercept and  $\mathbf{x}$  to those in (a).
- (c) Predict  $y_3$  from the regression in (a). Compare to the coefficient on  $\mathbf{d}$  in (b).

**A.2** A researcher fits  $\mathbf{y} = \mathbf{X}\beta + \mathbf{d}\delta + \mathbf{e}$ , where  $\mathbf{d}$  is a dummy for observation  $n$ . Prove that the  $n$ th residual is zero:  $\hat{e}_n = 0$ .

**A.3** In the same setup as A.2, prove that the coefficient on the dummy is

$$\hat{\delta} = y_n - \mathbf{x}'_n \hat{\beta}_{(-n)},$$

where  $\hat{\beta}_{(-n)}$  is the OLS estimate obtained by running the regression excluding observation  $n$ .

**A.4** Let  $h_{nn} = \mathbf{x}'_n(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_n$  denote the leverage of observation  $n$ , and let  $\hat{e}_n^{(\text{full})}$  be the  $n$ th residual from OLS of  $\mathbf{y}$  on  $\mathbf{X}$  only (no dummy).

- (a) Using the well-known identity  $\hat{e}_n^{(\text{full})} = (1 - h_{nn})(y_n - \mathbf{x}'_n \hat{\beta}_{(-n)})$ , express  $\hat{\delta}$  (from A.3) in terms of  $\hat{e}_n^{(\text{full})}$  and  $h_{nn}$ .
- (b) What happens to  $\hat{\delta}$  as  $h_{nn} \rightarrow 1$ ? Interpret.

**A.5** A researcher augments  $\mathbf{X}$  with unit dummies  $\mathbf{d}_1, \dots, \mathbf{d}_n$ , one for each of the  $n$  observations.

- (a) Why does this fail? What is the rank of the resulting design matrix?
- (b) Suppose she drops one dummy, leaving  $n - 1$  unit dummies plus  $\mathbf{X}$ . What are the fitted values? What is the  $R^2$ ?

## Concept B: OLS vs. ML variance estimators

**B.1** Under the homoskedastic normal model with  $n$  observations and  $k$  regressors, define

$$s_{\text{LS}}^2 = \frac{\mathbf{e}'\mathbf{e}}{n - k}, \quad s_{\text{ML}}^2 = \frac{\mathbf{e}'\mathbf{e}}{n}.$$

- (a) Show  $\mathbb{E}[s_{LS}^2] = \sigma^2$ . *Hint:  $\hat{\mathbf{e}}'\hat{\mathbf{e}} = \mathbf{e}'\mathbf{M}\mathbf{e}$  and  $\mathbb{E}[\mathbf{e}'\mathbf{M}\mathbf{e}] = \sigma^2 \text{tr}(\mathbf{M})$ .*  
 (b) Show  $\mathbb{E}[s_{ML}^2] = (n - k)\sigma^2/n$  and state the direction of the bias.

**B.2** With  $n = 20$  and  $k = 5$ :

- (a) Compute  $\mathbb{E}[s_{ML}^2]/\sigma^2$ . By what percentage does the ML estimator underestimate  $\sigma^2$  in expectation?  
 (b) Compute  $\text{Var}(s_{LS}^2)/\text{Var}(s_{ML}^2)$ . Which estimator has larger variance?

**B.3** Recall  $\text{Var}(s_{LS}^2) = 2\sigma^4/(n - k)$  and  $\text{Var}(s_{ML}^2) = 2(n - k)\sigma^4/n^2$ . Compute the mean squared error of each:

$$\text{MSE}(s) = \text{Var}(s) + \text{Bias}(s)^2.$$

For fixed  $n$  and  $k < n$ , which estimator has smaller MSE?

**B.4** Let  $n \rightarrow \infty$  with  $k$  fixed.

- (a) Show  $s_{LS}^2 \rightarrow \sigma^2$  in probability.  
 (b) Show  $s_{ML}^2 \rightarrow \sigma^2$  in probability.  
 (c) Both are consistent. Why does the finite-sample bias of  $s_{ML}^2$  not prevent consistency?

**B.5** In deriving  $\mathbb{E}[s_{LS}^2]$ , the key fact is  $\text{tr}(\mathbf{M}) = n - k$ . Prove this starting from  $\mathbf{M} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$  and using properties of the trace.

## Concept C: Sensitivity analysis

Throughout, use the Cinelli–Hazlett bound: for a single unobserved confounder  $Z$ ,

$$|\text{bias}| = \sqrt{\frac{R_{Y \sim Z|D, \mathbf{X}}^2 \cdot R_{D \sim Z|\mathbf{X}}^2}{1 - R_{D \sim Z|\mathbf{X}}^2}} \cdot \frac{s_e}{sd(\tilde{D})}.$$

**C.1** A researcher estimates  $\hat{\beta}_D = 2.0$  with  $s_e = 4$  and  $sd(\tilde{D}) = 2$ . Suppose an unobserved confounder has  $R_{Y \sim Z|D, \mathbf{X}}^2 = 0.04$  and  $R_{D \sim Z|\mathbf{X}}^2 = 0.04$ .

- (a) Compute the implied bias.  
 (b) Would the bias overturn the sign of  $\hat{\beta}_D$ ?

**C.2** The *robustness value*  $RV$  is the partial  $R^2$  value  $v$  such that if  $R_{Y \sim Z|D, \mathbf{X}}^2 = R_{D \sim Z|\mathbf{X}}^2 = v$ , the bias would equal  $|\hat{\beta}_D|$ .

- (a) Setting  $\text{bias} = |\hat{\beta}_D|$ , derive the equation

$$\frac{v^2}{1 - v} = \left( \frac{|\hat{\beta}_D| sd(\tilde{D})}{s_e} \right)^2.$$

- (b) For the numbers in C.1 ( $\hat{\beta}_D = 2$ ,  $s_e = 4$ ,  $sd(\tilde{D}) = 2$ ), solve for  $RV$ .
- C.3** Explain in 2–3 sentences why the bias formula depends on the *product* of the two partial  $R^2$  values, not on either one alone. What does this imply about a confounder that is highly correlated with  $D$  but uncorrelated with  $Y$  given  $D$ ?
- C.4** A researcher benchmarks against an observed covariate (incumbency) with  $R_{Y \sim \text{inc} | D, \mathbf{X}_{-\text{inc}}}^2 = 0.08$  and  $R_{D \sim \text{inc} | \mathbf{X}_{-\text{inc}}}^2 = 0.05$ . She finds  $\hat{\beta}_D = 1.2$  with  $s_e = 3$ ,  $sd(\tilde{D}) = 1$ .
- (a) Compute the bias a confounder “as strong as” incumbency would induce.
- (b) If the researcher argues the most likely unobserved confounder is *half* as strong as incumbency on each dimension, recompute the bias.
- C.5** Recall the textbook OVB decomposition  $\text{bias} = \text{imbalance} \times \text{impact}$ . In the Cinelli–Hazlett formula:
- (a) Which partial  $R^2$  captures imbalance? Which captures impact?
- (b) Why does the denominator  $1 - R_{D \sim Z | \mathbf{X}}^2$  appear? Give a one-sentence interpretation.

## Concept D: GLS under an AR(1) error process

Throughout, let  $\mathbf{e}_t = \rho \mathbf{e}_{t-1} + u_t$  with  $u_t$  iid  $(0, \sigma_u^2)$  and  $|\rho| < 1$ . The inverse transformation matrix is (showing the first few rows):

$$\mathbf{T}^{-1} = \begin{bmatrix} 1 & -\rho & 0 & \cdots \\ 0 & 1 & -\rho & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

**D.1** Let  $T = 3$ . Write out  $\mathbf{T}^{-1}$  explicitly as a  $3 \times 3$  matrix, then compute  $\mathbf{T} = (\mathbf{T}^{-1})^{-1}$ .

**D.2** For  $T = 3$ , compute  $\mathbf{T}\mathbf{T}'$  and verify that (up to the common factor  $1/(1 - \rho^2)$ ) it matches

$$\mathbf{\Omega} = \frac{1}{1 - \rho^2} \begin{bmatrix} 1 & \rho & \rho^2 \\ \rho & 1 & \rho \\ \rho^2 & \rho & 1 \end{bmatrix}.$$

(You may verify the diagonal entries and one off-diagonal entry.)

**D.3** The GLS estimator is  $\hat{\beta}_{\text{GLS}} = (\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{y}$ .

- (a) Define  $\tilde{\mathbf{y}} = \mathbf{T}^{-1}\mathbf{y}$  and  $\tilde{\mathbf{X}} = \mathbf{T}^{-1}\mathbf{X}$ . Show that OLS on  $(\tilde{\mathbf{y}}, \tilde{\mathbf{X}})$  returns  $\hat{\beta}_{\text{GLS}}$ .
- (b) What is the conditional variance of the transformed error  $\tilde{\mathbf{e}} = \mathbf{T}^{-1}\mathbf{e}$ ? (Up to a scalar.)

**D.4** Write out the transformed observations  $\tilde{y}_t$  in terms of  $y_t, y_{t-1}$ , and  $\rho$ , for  $t = 2, 3, \dots, T$ . Do the same for  $\tilde{x}_t$ . These are the “quasi-differenced” data used in Cochrane–Orcutt / Prais–Winsten.

**D.5** What does  $\mathbf{T}^{-1}$  reduce to when  $\rho = 0$ ? What about when  $\rho \rightarrow 1$ ? In each case, how does the GLS estimator relate to OLS?

## Concept E: Nonlinear CEFs and what linear regression recovers

**E.1** Suppose  $\mathbb{E}[Y | X] = (\gamma + \theta X)^{1/2}$  with  $\gamma, \theta > 0$ , and define  $u = Y - (\gamma + \theta X)^{1/2}$ . Let  $\text{Var}(u | X) = \sigma^2$  (homoskedastic).

- (a) Show  $\mathbb{E}[u | X] = 0$ .
- (b) Compute  $\mathbb{E}[Y^2 | X]$ .

**E.2** A researcher regresses  $y_i^2$  on  $x_i$  with an intercept:

$$y_i^2 = \alpha + \beta x_i + e_i.$$

Using the CEF from E.1:

- (a) What are the population values of  $\alpha$  and  $\beta$ ?
- (b) Does this regression identify  $\gamma$  and  $\theta$ ? Under what condition?

**E.3** Still using the CEF from E.1, construct the error  $e_i = y_i^2 - \alpha - \beta x_i$  (with  $\alpha, \beta$  as in E.2).

- (a) Show  $\mathbb{E}[e_i | x_i] = 0$ .
- (b) Is  $\text{Var}(e_i | x_i)$  constant in  $x_i$ ? Give a brief reason.

**E.4** Consider the log-linear CEF  $\mathbb{E}[Y | X] = \exp(a + bX)$  with  $Y > 0$ . A researcher regresses  $\log y_i$  on  $x_i$ :

$$\log y_i = a + bx_i + v_i.$$

- (a) Write  $\log y_i = \log \mathbb{E}[Y | X] + \log(Y/\mathbb{E}[Y | X])$ . Under what condition on  $Y$  does  $\mathbb{E}[\log(Y/\mathbb{E}[Y | X]) | X] = 0$ ?
- (b) In general, is  $\mathbb{E}[\log Y | X] = \log \mathbb{E}[Y | X]$ ? Name the inequality that decides the sign of the gap.

**E.5** A researcher is told that the CEF is quadratic:  $\mathbb{E}[Y | X] = \alpha + \beta X + \gamma X^2$ . She runs a *linear* regression  $y = a + bx + e$ .

- (a) Using the OVB formula, express the probability limit of the OLS slope  $b$  in terms of  $\beta$ ,  $\gamma$ , and the regression of  $x^2$  on  $x$ .
- (b) Under what distributional symmetry on  $X$  does the OLS slope still consistently estimate  $\beta$ ?

## Solutions

**A.1.** (a) With two points (1, 2) and (2, 5), the line through them is  $y = -1 + 3x$ , so  $\hat{\alpha}_{(-3)} = -1$ ,  $\hat{\beta}_{(-3)} = 3$ . (b) In the three-observation regression with a unit dummy for observation 3, the intercept and slope on  $\mathbf{x}$  remain  $-1$  and  $3$ , identical to (a). (c) The prediction from (a) at  $x = 3$  is  $-1 + 3(3) = 8$ . The actual  $y_3 = 4$ , so  $y_3 - \hat{y}_3^{(\text{out of sample})} = 4 - 8 = -4$ , which equals  $\hat{\delta}$  (the coefficient on  $\mathbf{d}$ ).

**A.2.** By FWL, the regression with  $\mathbf{d}$  added is equivalent to regressing  $\mathbf{M}_X \mathbf{y}$  on  $\mathbf{M}_X \mathbf{d}$ , where  $\mathbf{M}_X$  annihilates  $\mathbf{X}$ . The residuals of the full regression coincide with the residuals from this one-variable regression. Fitted values at observation  $n$ : since  $\mathbf{d}$  is 1 only at position  $n$ , OLS will always be able to perfectly fit observation  $n$  using  $\delta$  (an unconstrained scalar). More directly: the partitioned normal equations force  $\mathbf{d}'\hat{\mathbf{e}} = \hat{e}_n = 0$ .

**A.3.** By FWL,  $\hat{\delta} = (\mathbf{d}'\mathbf{M}_X\mathbf{d})^{-1}\mathbf{d}'\mathbf{M}_X\mathbf{y}$ . Write  $\mathbf{M}_X\mathbf{y} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{(-n)}$  (adjustment for obs  $n$ ); a cleaner route is to note from A.2 that  $\hat{e}_n = 0$ , i.e.  $y_n - \mathbf{x}'_n\hat{\boldsymbol{\beta}} - \hat{\delta} = 0$ , so  $\hat{\delta} = y_n - \mathbf{x}'_n\hat{\boldsymbol{\beta}}$ . But by the FWL logic (coefficients on  $\mathbf{X}$  in the full regression equal those from leaving out obs  $n$ ),  $\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}_{(-n)}$ , so  $\hat{\delta} = y_n - \mathbf{x}'_n\hat{\boldsymbol{\beta}}_{(-n)}$ .

**A.4.** (a) From A.3,  $\hat{\delta} = y_n - \mathbf{x}'_n\hat{\boldsymbol{\beta}}_{(-n)}$ . Using the identity  $\hat{e}_n^{(\text{full})} = (1 - h_{nn})(y_n - \mathbf{x}'_n\hat{\boldsymbol{\beta}}_{(-n)})$ , solve:  $y_n - \mathbf{x}'_n\hat{\boldsymbol{\beta}}_{(-n)} = \hat{e}_n^{(\text{full})}/(1 - h_{nn})$ , so  $\hat{\delta} = \hat{e}_n^{(\text{full})}/(1 - h_{nn})$ . (b) As  $h_{nn} \rightarrow 1$  the denominator vanishes and  $\hat{\delta}$  blows up. High-leverage points are “almost” determined by  $\mathbf{X}$  alone — the small residual from the full regression is amplified by  $1/(1 - h_{nn})$  when one backs it out to the LOO fit. This is the Cook’s-distance / leverage mechanism.

**A.5.** (a) With  $n$  unit dummies, their sum is the vector of ones, which duplicates (some linear combination of) the intercept column already in  $\mathbf{X}$ . Rank is at most  $n$ , and the augmented design is exactly singular, so  $(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}$  does not exist. (b) With  $n - 1$  dummies,  $\mathbf{X}$  becomes a saturated design for the  $n$  observations: the column span is  $\mathbb{R}^n$ . Fitted values equal  $\mathbf{y}$  exactly, all residuals are 0, and  $R^2 = 1$ .

**B.1.** (a)  $\mathbb{E}[\hat{\mathbf{e}}'\hat{\mathbf{e}}] = \mathbb{E}[\mathbf{e}'\mathbf{M}\mathbf{e}] = \sigma^2 \text{tr}(\mathbf{M}) = \sigma^2(n - k)$ , so  $\mathbb{E}[s_{\text{LS}}^2] = \sigma^2(n - k)/(n - k) = \sigma^2$ . (b)  $\mathbb{E}[s_{\text{ML}}^2] = \sigma^2(n - k)/n < \sigma^2$ ; the bias is downward.

**B.2.** (a)  $\mathbb{E}[s_{\text{ML}}^2]/\sigma^2 = (n - k)/n = 15/20 = 0.75$ , a 25% downward bias. (b)  $\text{Var}(s_{\text{LS}}^2)/\text{Var}(s_{\text{ML}}^2) = [2\sigma^4/(n - k)]/[2(n - k)\sigma^4/n^2] = n^2/(n - k)^2 = 400/225 \approx 1.78$ . The LS estimator has about 78% higher variance.

**B.3.**  $\text{MSE}(s_{\text{LS}}^2) = 2\sigma^4/(n - k)$ . For  $s_{\text{ML}}^2$ : bias  $= -k\sigma^2/n$ , so bias<sup>2</sup>  $= k^2\sigma^4/n^2$ , and  $\text{MSE}(s_{\text{ML}}^2) = 2(n - k)\sigma^4/n^2 + k^2\sigma^4/n^2 = \sigma^4(2n - 2k + k^2)/n^2$ . Compute the ratio or sign of the difference:

$$\text{MSE}(s_{\text{LS}}^2) - \text{MSE}(s_{\text{ML}}^2) = \sigma^4 \left[ \frac{2}{n - k} - \frac{2n - 2k + k^2}{n^2} \right].$$

For  $n = 20, k = 5$ :  $2/15 - (40 - 10 + 25)/400 = 0.1333 - 0.1375 < 0$ , so  $s_{\text{LS}}^2$  has slightly smaller MSE. More generally, *neither estimator dominates uniformly*: the ranking depends on  $(n, k)$  in a non-monotone way.  $s_{\text{ML}}^2$ 's bias<sup>2</sup> grows like  $k^2\sigma^4/n^2$  while  $s_{\text{LS}}^2$ 's variance blows up like  $1/(n - k)$  as  $k$  approaches  $n$ , so  $s_{\text{ML}}^2$  wins both when  $k$  is very small (negligible bias, lower variance) and when  $k$  is close to  $n$  (LS variance explodes). The bias-variance trade-off is genuine, not monotone in  $k/n$ .

**B.4.** (a)  $\mathbb{E}[s_{LS}^2] = \sigma^2$  and  $\text{Var}(s_{LS}^2) = 2\sigma^4/(n-k) \rightarrow 0$ , so by Chebyshev  $s_{LS}^2 \rightarrow \sigma^2$  in probability. (b)  $s_{ML}^2 = \frac{n-k}{n}s_{LS}^2 \rightarrow 1 \cdot \sigma^2 = \sigma^2$  by Slutsky, since  $(n-k)/n \rightarrow 1$ . (c) Consistency requires only that bias and variance both vanish as  $n \rightarrow \infty$ . Here the bias  $-k\sigma^2/n \rightarrow 0$  and the variance  $\rightarrow 0$ , so both estimators converge in probability to  $\sigma^2$ . A fixed finite-sample bias is not the same as a non-vanishing asymptotic bias.

**B.5.**  $\text{tr}(\mathbf{M}) = \text{tr}(\mathbf{I}_n) - \text{tr}(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') = n - \text{tr}((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}) = n - \text{tr}(\mathbf{I}_k) = n - k$ . The cyclic property of the trace is what makes the second equality work.

**C.1.** (a)  $|\text{bias}| = \sqrt{(0.04)(0.04)/(1-0.04)} \cdot (4/2) = \sqrt{0.0016/0.96} \cdot 2 = \sqrt{0.001667} \cdot 2 \approx 0.0408 \cdot 2 = 0.0816$ . (b) With  $\hat{\beta}_D = 2$  and a worst-case bias of  $\approx 0.08$ , the sign and qualitative conclusion are unchanged.

**C.2.** (a) Set  $|\text{bias}|^2 = \hat{\beta}_D^2 \cdot \frac{v \cdot v}{1-v} \cdot \frac{s_e^2}{sd(\tilde{D})^2} = \hat{\beta}_D^2$ , which rearranges to  $v^2/(1-v) = (\hat{\beta}_D sd(\tilde{D})/s_e)^2$ . (b) RHS  $= (2 \cdot 2/4)^2 = 1$ , so  $v^2 = 1 - v$ , i.e.  $v^2 + v - 1 = 0$ . The positive root is  $v = (-1 + \sqrt{5})/2 \approx 0.618$ . A confounder would need to explain  $\sim 62\%$  of the residual variation in  $Y$  and  $D$  to overturn the estimate — very hard to satisfy.

**C.3.** Bias requires  $Z$  to be related to *both*  $D$  (to create imbalance) and  $Y$  (to matter for the outcome). If  $Z$  strongly predicts  $D$  but is unrelated to  $Y$  conditional on  $D$ , the outcome-side partial  $R^2$  is zero and the product is zero — no bias. This is why “strong predictors of treatment” are not automatically threats: they matter only if they also drive the outcome.

**C.4.** (a)  $|\text{bias}| = \sqrt{(0.08)(0.05)/(1-0.05)} \cdot (3/1) = \sqrt{0.004/0.95} \cdot 3 = \sqrt{0.00421} \cdot 3 \approx 0.0649 \cdot 3 \approx 0.195$ . About 16% of  $\hat{\beta}_D = 1.2$ , so the estimate would survive qualitatively. (b) Half as strong on each dimension:  $R_{Y \sim Z}^2 = 0.04$ ,  $R_{D \sim Z}^2 = 0.025$ . Bias  $= \sqrt{(0.04)(0.025)/(1-0.025)} \cdot 3 = \sqrt{0.001026} \cdot 3 \approx 0.0320 \cdot 3 \approx 0.096$ , about 8% of the estimate.

**C.5.** (a)  $R_{D \sim Z|\mathbf{X}}^2 = \text{imbalance}$  (how much  $Z$  differs across treatment conditions after netting out  $\mathbf{X}$ ).  $R_{Y \sim Z|D, \mathbf{X}}^2 = \text{impact}$  (how much  $Z$  moves  $Y$  once  $D$  and  $\mathbf{X}$  are held fixed). (b) The denominator  $1 - R_{D \sim Z|\mathbf{X}}^2$  inflates the bias when  $Z$  is a near-perfect predictor of  $D$ : there is less residual variation in  $\tilde{D}$  with which to identify  $\beta_D$ , so the same correlation with  $Y$  translates into a larger bias per unit of residual treatment variation.

**D.1.** For  $T = 3$ ,

$$\mathbf{T}^{-1} = \begin{bmatrix} 1 & -\rho & 0 \\ 0 & 1 & -\rho \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{T} = \begin{bmatrix} 1 & \rho & \rho^2 \\ 0 & 1 & \rho \\ 0 & 0 & 1 \end{bmatrix}.$$

Check: multiply, upper-triangular with ones on the diagonal and zero elsewhere.

**D.2.** Actually  $\mathbf{T}\mathbf{T}' = \Omega$  holds only in the limiting / stationary sense, where the formula has the  $1/(1-\rho^2)$  prefactor. For the truncated  $3 \times 3$  case,  $\mathbf{T}\mathbf{T}' = \begin{bmatrix} 1+\rho^2+\rho^4 & \rho+\rho^3 & \rho^2 \\ \rho+\rho^3 & 1+\rho^2 & \rho \\ \rho^2 & \rho & 1 \end{bmatrix}$ . The  $(3,3)$  entry is 1 (as in  $\Omega$  times  $1-\rho^2$ ). The  $(2,3)$  entry is  $\rho$  (matches  $\rho \cdot (1-\rho^2)$  only for the initial condition without stationarity).

adjustment). The full stationary identity  $\mathbf{T}\mathbf{T}' = \Omega$  requires taking  $T \rightarrow \infty$  or modifying the first row of  $\mathbf{T}^{-1}$  to  $(\sqrt{1-\rho^2}, 0, \dots)$  as in Prais–Winsten. The problem flags this subtlety deliberately.

**D.3.** (a)  $\tilde{\mathbf{y}} = \mathbf{T}^{-1}\mathbf{y}$ ,  $\tilde{\mathbf{X}} = \mathbf{T}^{-1}\mathbf{X}$ . OLS on the transformed data:  $(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}'\tilde{\mathbf{y}} = (\mathbf{X}'(\mathbf{T}^{-1})'\mathbf{T}^{-1}\mathbf{X})^{-1}\mathbf{X}'(\mathbf{T}^{-1})'\mathbf{T}^{-1}\mathbf{y}$ . Since  $(\mathbf{T}^{-1})'\mathbf{T}^{-1} = (\mathbf{T}\mathbf{T}')^{-1} = \Omega^{-1}$ , this is  $(\mathbf{X}'\Omega^{-1}\mathbf{X})^{-1}\mathbf{X}'\Omega^{-1}\mathbf{y} = \hat{\beta}_{\text{GLS}}$ . (b)  $\text{Var}(\tilde{\mathbf{e}}) = \mathbf{T}^{-1}\text{Var}(\mathbf{e})(\mathbf{T}^{-1})' = \sigma^2\mathbf{T}^{-1}\Omega(\mathbf{T}^{-1})' = \sigma^2\mathbf{I}$  (in the stationary case, up to the first-observation adjustment). Transformed errors are homoskedastic and uncorrelated, which is why OLS on the transformed data is efficient.

**D.4.** For  $t \geq 2$ :  $\tilde{y}_t = y_t - \rho y_{t-1}$  and  $\tilde{x}_t = x_t - \rho x_{t-1}$ . These are quasi-differences. The first observation is either dropped (Cochrane–Orcutt) or rescaled by  $\sqrt{1-\rho^2}$  (Prais–Winsten) to preserve the stationary covariance structure.

**D.5.** When  $\rho = 0$ :  $\mathbf{T}^{-1} = \mathbf{I}$ , no transformation needed, and GLS = OLS (errors are already iid). When  $\rho \rightarrow 1$ :  $\mathbf{T}^{-1}$  becomes the first-difference operator (ignoring the first row), so  $\tilde{y}_t = \Delta y_t$ . GLS then runs OLS on first differences — the standard remedy for a random walk in errors.

**E.1.** (a)  $\mathbb{E}[u | X] = \mathbb{E}[Y | X] - (\gamma + \theta X)^{1/2} = 0$  by the definition of the CEF. (b)  $\mathbb{E}[Y^2 | X] = \text{Var}(Y | X) + \mathbb{E}[Y | X]^2 = \sigma^2 + (\gamma + \theta X)$ .

**E.2.** (a) The population projection coefficients satisfy  $\mathbb{E}[Y^2 | X] = \alpha + \beta X$ . From E.1(b),  $\mathbb{E}[Y^2 | X] = (\gamma + \sigma^2) + \theta X$ , so  $\alpha = \gamma + \sigma^2$  and  $\beta = \theta$ . (b)  $\theta$  is identified directly as  $\beta$ . But  $\gamma$  is confounded with  $\sigma^2$ : we can recover  $\gamma$  only if  $\sigma^2$  is known separately (or if homoskedasticity is imposed and  $\sigma^2$  is estimated from residuals). Otherwise only  $\gamma + \sigma^2$  is estimable from  $\alpha$ .

**E.3.** (a) Write  $y_i^2 = \alpha + \beta x_i + e_i$  with  $\alpha = \gamma + \sigma^2$ ,  $\beta = \theta$ . Then  $e_i = y_i^2 - \mathbb{E}[Y^2 | X = x_i]$ , so  $\mathbb{E}[e_i | x_i] = 0$  by definition. (b) Generally no.  $\text{Var}(y_i^2 | x_i)$  depends on higher moments of  $Y$  given  $X$ , which themselves depend on  $x_i$  through the mean  $(\gamma + \theta x_i)^{1/2}$ . For instance, if  $Y | X$  is normal,  $\text{Var}(Y^2 | X) = 4\sigma^2 \mathbb{E}[Y | X]^2 + 2\sigma^4$ , which grows with  $x_i$ . The linear regression in  $y^2$  is typically heteroskedastic.

**E.4.** (a) Write  $\log Y = \log \mathbb{E}[Y | X] + \log(Y/\mathbb{E}[Y | X]) = a + bX + v$ . Then  $\mathbb{E}[v | X] = 0$  iff  $\mathbb{E}[\log(Y/\mathbb{E}[Y | X]) | X] = 0$ . This holds, for example, if  $Y/\mathbb{E}[Y | X]$  has a distribution that does not depend on  $X$  (sometimes called a multiplicative error model:  $Y = e^{a+bX} \cdot \eta$  with  $\mathbb{E}[\log \eta] = 0$ ). (b) No. By Jensen's inequality,  $\mathbb{E}[\log Y | X] \leq \log \mathbb{E}[Y | X]$  (since log is concave), with equality only if  $Y$  is degenerate given  $X$ . The gap is  $-\text{Var}(\log Y | X)/2$  to first order for small noise.

**E.5.** (a) Let  $\tilde{\delta} = \text{Cov}(X, X^2)/\text{Var}(X)$  denote the population projection slope of  $X^2$  on  $X$ . Then  $\text{plim } b = \beta + \gamma\tilde{\delta}$ . (b) The OLS slope is consistent for  $\beta$  when  $\tilde{\delta} = 0$ , i.e.  $\text{Cov}(X, X^2) = 0$ . For  $X$  symmetric about 0,  $\mathbb{E}[X^3] = 0$  and  $\mathbb{E}[X]\mathbb{E}[X^2] = 0$ , so  $\text{Cov}(X, X^2) = 0$  and  $b \rightarrow \beta$ . *Symmetry about 0 is essential*: if  $X$  is symmetric about  $\mu \neq 0$ , write  $X = \mu + Z$  with  $Z$  symmetric about zero, then  $\text{Cov}(X, X^2) = \text{Cov}(Z, Z^2 + 2\mu Z) = 2\mu\sigma^2 \neq 0$ , so  $b \rightarrow \beta + 2\gamma\mu$ . This is why zero-mean (or more generally, odd-symmetric)  $X$  is required for linear regression to be “accidentally” consistent when the CEF is quadratic.